

Financial Engineering Analysis

1. Single Payment Compound-Amount Factor[SPCAF]:-

Example 1: A contractor wishes to set up a revolving line of credit at the bank to handle his cashflow during the construction of a project. He believes that he needs to borrow 12,000Birr with which to set up the account, and that he can obtain the money at 1.45% per month. If he pays back the loan and accumulated interest after 8 months, how much will he/she have to pay back?

Solution:-

$$\begin{aligned} F &= P * [1 + i]^N, = \text{Notation: } -12,000 * \left[\frac{F}{P}, 1.45\%, 8\right] \\ &= 12,000 * \left[1 + 1. \frac{45}{100}\right]^8 \\ &= 12,000 * 1.0145^8 \\ &= 13,464.73 \end{aligned}$$

Therefore, the amount of interest will be: $13,464.73 - 12,000 = 1,464.73 \text{ ETB}$

2. Single Payment Present-Worth Factor [SPPWF]:

Example 2: A construction company wants to set aside enough money today in an interest-bearing account in order to have 100,000Birr five years from now for the purchase of a replacement piece of equipment. If the company can receive 8% interest on its investment, how much should be set aside now to collect the **100,000 Birr** five years from now?

Solution:-

$$\begin{aligned} P &= F * \left\{ \frac{1}{[1 + i]^N} \right\} \\ &= \text{Notation: } 100,000 * \left[\frac{P}{F}, 8\%, 5\right] \\ P &= 100,000 * \left\{ \frac{1}{[1 + 0.08]^5} \right\} \\ &= \frac{100,000}{1.4693} \\ &= 68,058.32 \text{ ETB} \end{aligned}$$

3. Uniform Series Compound-Amount Factor [USCAF]:

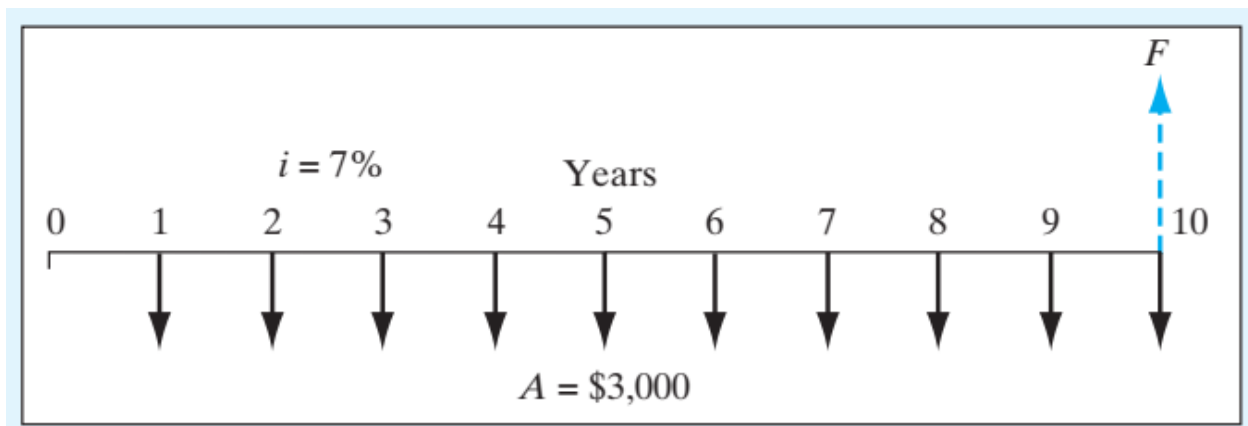
Example 3:- Suppose you make an annual contribution of \$3,000 to your savings account at the end of each year for 10 years. If the account earns 7% interest annually, how much can be withdrawn at the end of 10 years

Solution:-

Given: $A = 3000$ USD, Interest per year, $i = 7\%$, Interest Period $N = 10$ years and Find: F .

$$\begin{aligned} F &= 3000 * \left\{ \frac{(1+0.07)^{10}-1}{0.07} \right\} \\ &= 3000 \left\{ \frac{F}{A}, 7\%, 10 \right\} \text{ Notation} \\ &= 3000 * \{13.8164\} \\ &= \$41,449.20 \end{aligned}$$

CFD:- Cash Flow Diagram:-



4. Uniform [Equal payment] Series Sinking-Fund Factor [USSFF]:

Example 4:- What uniform monthly amount should you deposit in your savings account at the end of each month for the following **10 months in order to accumulate \$75,000 at the time of the 10th deposit?** Assume that the interest rate you can earn is **4% per month** and the first deposit will be made next month.

Solution:

Given: $F = 75,000$, $i = 4\%$ per month, $N = 10$ months.

Find: $A = ?$

$$A = F \left(\frac{i}{(1+i)^n - 1} \right) \quad A = \left(\frac{A}{F}, i, n \right)$$

Notation

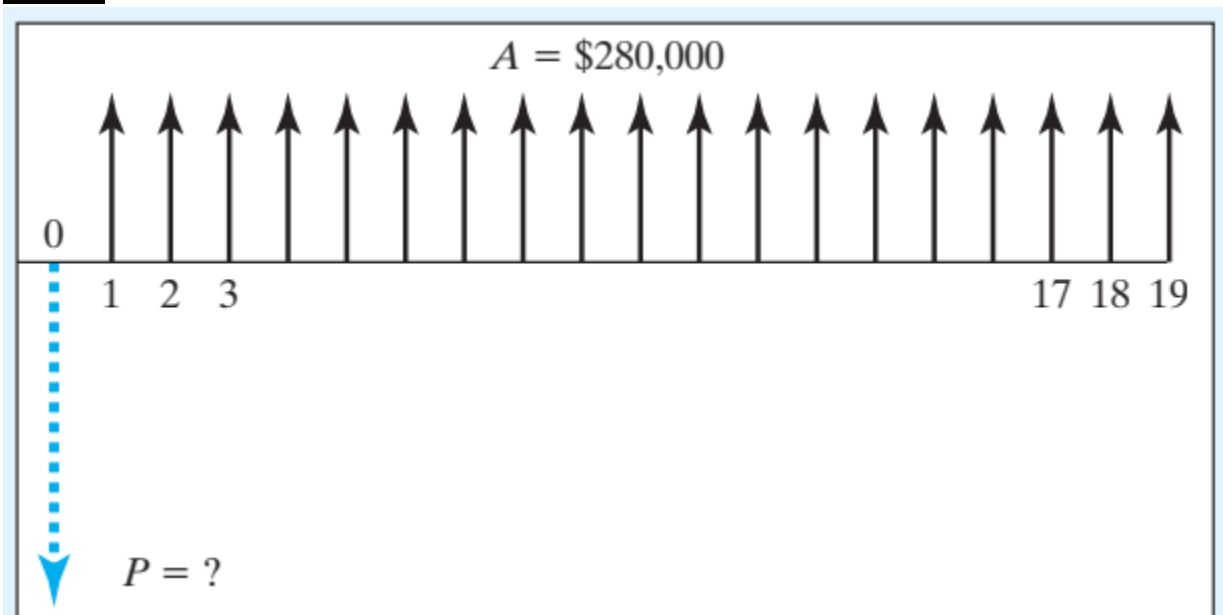
$$A = F \left[\frac{i}{(1+i)^N - 1} \right] = 75,000 \left[\frac{0.04}{(1.04)^{10} - 1} \right] = \$6,247$$

5. Uniform [Equal payment] Series Present-Worth Factor [USPWF]:

Example 5:- Suppose that Outing were able to find an investor who was willing to buy her lottery ticket for \$2 million. Recall that after an initial gross payment of \$283,770, Outing would be paid 19 annual gross checks of \$280,000. (See Figure 3.26.) If she could invest her money at 8% interest, what would be the fair amount to trade her 19 future lottery receipts? (Note that she already cashed in \$283,770 after winning the lottery, so she is giving up 19 future lottery checks in the amount of \$280,000.)

Solution:-

CFD:-



$$A = P \left(\frac{i(1+i)^n}{(1+i)^n - 1} \right)$$

$$A = \left(\frac{A}{P}, i, n \right)$$

Notation

SOLUTION

Given: $i = 8\%$ per year, $A = \$280,000$, and $N = 19$ years.

Find: P .

- Using interest factor:

$$\begin{aligned} P &= \$280,000(P/A, 8\%, 19) = \$280,000(9.6036) \\ &= \$2,689,008. \end{aligned}$$

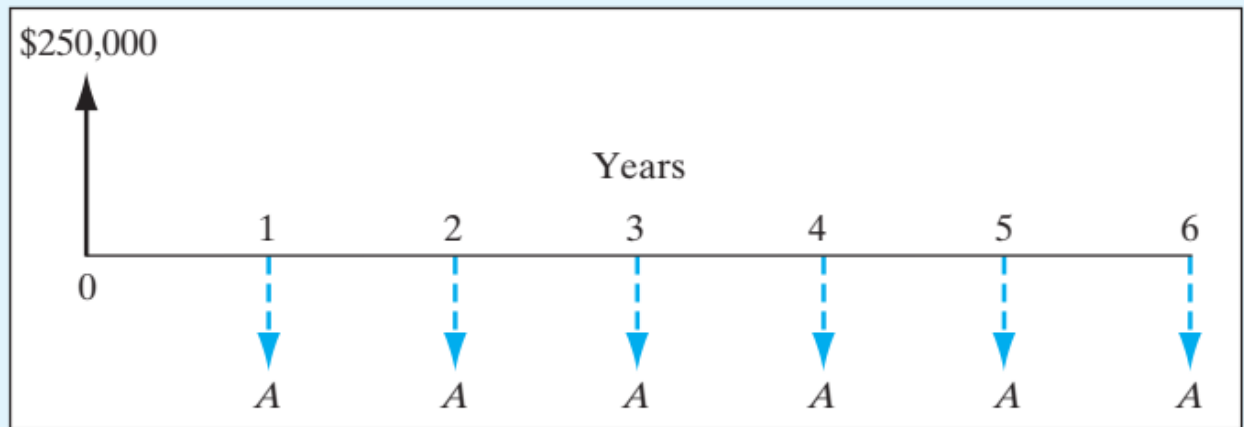
6. Uniform [Equal payment] Series Capital-Recovery Factor [USCRF]:

Example 6:- A small biotechnology firm, has borrowed \$250,000 to purchase laboratory equipment for gene splicing. The loan carries an interest rate of 8% per year and is to be repaid in equal installments over the next six years. Compute the amount of the annual installment

Solution:

Given: Present Value, $P = \$250,000$, Interest per Year, $I = 8\%$, and Interest Period, $N = 6$ years and Find: - $A = ?$ as shown in the CFD figure

CFD:-



$$A = P \left(\frac{i(1+i)^n}{(1+i)^n - 1} \right)$$

$$A = \left(\frac{A}{P}, i, n \right)$$

write the Notation

Given: $P = \$250,000$, $i = 8\%$ per year, and $N = 6$ years.
Find: A .

$$\begin{aligned} A &= \$250,000(A/P, 8\%, 6) \\ &= \$250,000(0.2163) \\ &= \$54,075. \end{aligned}$$

Economic Equivalence

Example:

Suppose you are offered the alternative of receiving either \$3,000 at the end of five years or P dollars today. There is no question that the \$3,000 will be paid in full (no risk). Because you have no current need for the money, you would deposit the P dollars in an account that pays 8% interest. What value of P would make you indifferent to your choice between P dollars today and the promise of \$3,000 at the end of five years?

Our job is to determine the present amount that is economically equivalent to \$3,000 in five years, given the investment potential of 8% per year.

SOLUTION

Given: $F = \$3,000$, $N = 5$ years, and $i = 8\%$ per year.

Find: P .

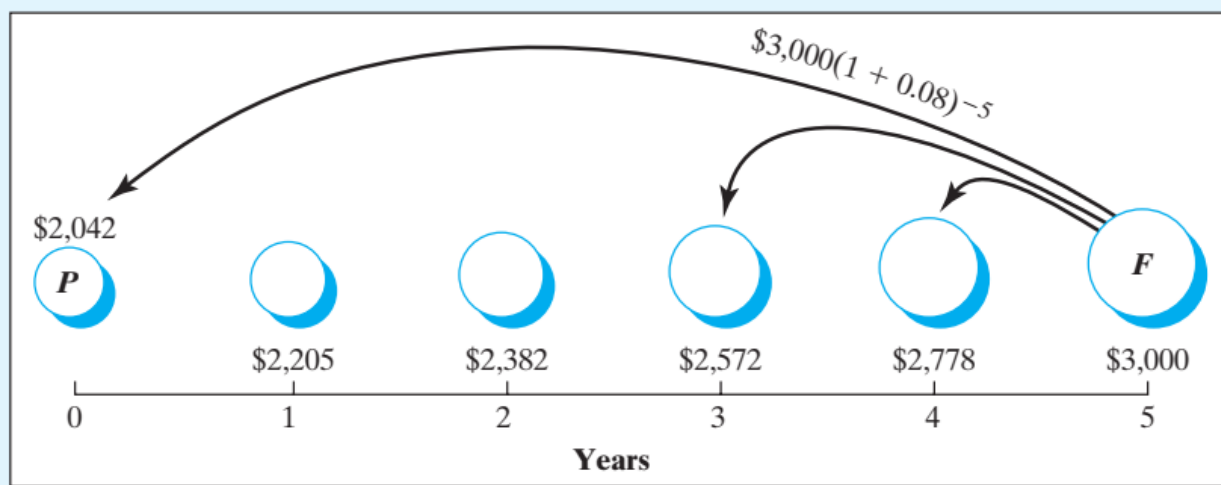
Equation: Eq. (3.3), $F = P(1 + i)^N$.

Rearranging terms to solve for P gives

$$P = \frac{F}{(1 + i)^N}.$$

Substituting yields

$$P = \frac{\$3,000}{(1 + 0.08)^5} = \$2,042.$$



Comments:- In this example, it is clear that if P is anything less than \$2,042, you would prefer the promise of \$3,000 in five years to P dollars today; if P is greater than \$2,042, you would prefer P . As you may have already guessed, at a lower interest rate, P must be higher to be equivalent to the future amount. For example, at $i = 4\%$, $P = \$2,466$.

Cash Flow Diagram